

Table 10. An ARC example and its 4 associated counterfactuals.

Prompt / Counterfactual	Text	Correct Completion
Original Prompt	<i>Question: How does a tiger get stripes?</i> <i>A. from its environment</i> <i>B. from its food</i> <i>C. from its offspring</i> <i>D. from its parents</i> <i>Answer:</i>	D
Answer Position	<i>Question: How does a tiger get stripes?</i> <i>A. from its food</i> <i>B. from its parents</i> <i>C. from its environment</i> <i>D. from its offspring</i> <i>Answer:</i>	B
Symbol	<i>Question: How does a tiger get stripes?</i> <i>1. from its environment</i> <i>2. from its food</i> <i>3. from its offspring</i> <i>4. from its parents</i> <i>Answer:</i>	4
Random Letter	<i>Question: How does a tiger get stripes?</i> <i>D. from its environment</i> <i>H. from its food</i> <i>M. from its offspring</i> <i>E. from its parents</i> <i>Answer:</i>	E
Answer Position + Random Letter	<i>Question: How does a tiger get stripes?</i> <i>D. from its food</i> <i>H. from its parents</i> <i>M. from its environment</i> <i>E. from its offspring</i> <i>Answer:</i>	H

C.5. Model Performance

For all tasks, we report accuracy given greedy generations in Table 1. For tasks that involve selecting between a fixed set of answer choices (IOI, MCQA, and ARC), we additionally report ranked-choice accuracy in Table 11. Ranked-choice scoring computes a model’s prediction as the token that is assigned the highest probability within the set of answer choices; this is an upper bound on greedy generation performance. Ranked-choice scoring is more in line with the metric m used for circuit localization (§3.1); greedy scoring is more in line with the prerequisite of causal variable localization (§4).

In Table 12 and Table 13 we report the results for each counterfactual type in the MCQA and ARC tasks, respectively. For the IOI and Arithmetic tasks, we found that due to the counterfactual format, all counterfactual types lead to the same performance as the original prompts (except for the “ABC” counterfactual in IOI, which has no correct answer).

C.6. Leaderboard

Here, we present screenshots of the MIB leaderboards. The leaderboards for both tracks are hosted on the same webpage; they are in separate tabs. The circuit localization track’s leaderboard can be viewed in Figure 5. It has two tabs: one for CPR and one for CMD. Each row displays an Average score, which is a macroaverage of all scores in the row. Each row also

MIB: A Mechanistic Interpretability Benchmark

[Circuit Localization](#) [Causal Variable Localization](#) [Submit](#)

[CPR](#) [CMD](#)

Filtering Options

Use the dropdown menus below to filter results by specific tasks or models. You can combine filters to see specific task-model combinations.

View tasks:

☒ IOI ☒ MCQA ☐ Arithmetic ☒ ARC

View models:

☒ GPT-2 ☐ Qwen-2.5 ☐ Gemma-2 ☒ Llama-3.1

Search

Separate multiple queries with ','.

Method	IOI - GPT-2	IOI - Llama-3.1	MCQA - Llama-3.1	ARC (Easy) - Llama-3.1	ARC (Challenge) - Llama-3.1	Average	Score
EAP-IG-inputs (CF)	1.89	2.04	1.05	1.05	0.3	1.89	0.86
EAP-IG-activations (CF)	1.84	1.33	0.83	0.25	0.26	1.84	0.84
EAP (CF)	1.26	0.87	1.06	0.8	0.26	1.26	0.63
NAP-IG-inputs (CF)	0.69	0.42	1.84	0.26	0.26	0.69	0.35
IFR	0.57	0.11	0.51	0.32	0.44	0.57	0.3
EAP (mean)	0.28	0.94	0.13	0.34	0.31	0.28	0.19
NAP (CF)	0.27	0.25	1.69	0.26	0.26	0.27	0.19
EAP (OA)	1.07	-	-	-	-	-	-
UGS	0.97	-	-	-	-	-	-
EActP (CF)	2.39	-	-	-	-	-	-

Figure 5. Leaderboard for the circuit localization track.

The leaderboards for each track of the Mechanistic Interpretability Benchmark.

[Circuit Localization](#) [Causal Variable Localization](#) [Submit](#)

[Highest View](#) [Averaged View](#)

Filtering Options

Use the dropdown menus below to filter results by specific tasks or models. You can combine filters to see specific task-model combinations.

View tasks:

☐ IOI ☐ MCQA ☒ ARC (Easy) ☐ RAVEL

View models:

☐ GPT-2 ☐ Qwen-2.5 ☒ Gemma-2 ☒ Llama-3.1

Search

Separate multiple queries with ','.

Method	ARC (Easy) - Llama-3.1 - Answer Pointer	ARC (Easy) - Gemma-2 - Answer	ARC (Easy) - Llama-3.1 - Answer	ARC (Easy) - Gemma-2 - Answer Pointer
DAS	0.836	0.941	0.988	0.876
DBM	0.818	0.986	0.998	0.802
full_vector	0.719	1	1	0.735
DBM+PCA	0.829	0.983	0.998	0.809
DBM+SAE	0.669	0.887	0.942	0.7

Figure 6. Leaderboard for the causal variable localization track.

Table 11. Model performance for all models on the public test split of each analyzed task (0-shot) with ranked-choice accuracy.

	IOI	MCQA	ARC	
			(E)	(C)
Llama-3.1 8B	1.00	0.92	0.93	0.79
Gemma-2 2B	1.00	1.00	0.79	0.60
Qwen-2.5 0.5B	1.00	1.00	0.73	0.58
GPT-2 Small	1.00	0.30	0.23	0.23

Table 12. Model accuracy (0-shot, greedy generation) on MCQA counterfactuals. N = Noun, C = Color, AP = Answer Position, S = Symbol, RL = Random Letter

Model	N	C	N+C	AP	S	RL	AP+RL	AP + S	AP + C
Llama 3.1-8B	0.70	0.72	0.96	0.94	0.94	0.90	0.98	0.98	0.72
Gemma 2-2B	0.92	1.00	1.00	1.00	0.98	0.70	0.74	1.00	0.98
Qwen 2.5-0.5B	1.00	1.00	1.00	1.00	1.00	0.52	0.60	1.00	0.98
GPT2-Small	0.00	0.06	0.00	0.02	0.00	0.00	0.00	0.00	0.02

displays a Score column, which is an average of the sigmoid of each score; we apply a sigmoid because each faithfulness value exists in a separate scale, where the lower and upper bounds may be different. Thus, to prevent any one column from dominating the score, we normalize each score to a $[0, 1]$ range by applying a sigmoid to each CPR or CMD value before averaging. Users may filter rows based on model name and/or task name. After filtering, the Average and Score columns are recomputed dynamically. This allows users to compare performance at varying levels of granularity, or in cases where some methods are only tractable to run for a subset of the task/model combinations.

The causal variable localization track’s leaderboard can be viewed in Figure 6. It displays results aggregated across all layers and token positions, *and* across counterfactual types. As in the circuit localization track, the Average is recomputed dynamically after filtering.

Our leaderboard will accept user submissions. To submit to the circuit localization track, a user must supply either (i) importance scores $\in \mathbb{R}$ on each node or edge, or (ii) 9 circuits of different sizes with membership in $\mathcal{C}$ given as Booleans. Recall from §3 that we use circuits containing varying percentages of edges; we will enforce this as a submission requirement for the circuit localization track. The smallest circuit can contain k edges in the model, where $\frac{|C_k|}{|N|} \leq 0.001$. The second-smallest can contain any proportion of edges $k \leq 0.2$; the largest can contain $k \leq 0.5$, and so on.

For the causal variable localization track, a user must provide an invertible featurizer function $\mathcal{F}$ and token position functions specifying where in an input to apply them. These must be provided as Python scripts. The trained featurizer, inverse featurizer, and token indices must also be provided for each task/model combination. We will evaluate whether interchange interventions on the features and the variable result in the same behavior for each counterfactual dataset and average the results across them.

D. Details on Circuit Localization Track

D.1. Methods

Mapping from scores to circuits. All of the techniques that we benchmark produce a set of scores, which must then be mapped to circuits. To find a circuit with n edges, we can simply take the top- n edges by score—what we call the *top- n* method for constructing circuits. However, this approach can often result in a circuit without an end-to-end pathway from inputs to outputs. Alternatively, we can perform a *greedy search* starting from the logits as follows. Let our circuit $C = (V_C, V_E) = (\{\text{logits}\}, \emptyset)$. Then for $i = 1, \dots, n$ add the highest- $\hat{I}\hat{E}$ edge connected to V_C that is not currently in V_E , to V_E ; add its parent to V_C . For simplicity, we use top- n , except in cases where it tends to not work well; in practice, greedy circuit construction is only needed for good performance when using information flow routes.

When deciding which components to use, we can either add to the circuit the edges with the highest score. Or, we can first